



National
Qualifications
2024

2024 Biology

National 5

Question Paper Finalised Marking Instructions

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Marking Instructions for each question

Section 1

Question	Response	Mark
1.	C	1
2.	C	1
3.	D	1
4.	A	1
5.	C	1
6.	D	1
7.	B	1
8.	B	1
9.	C	1
10.	A	1
11.	D	1
12.	D	1
13.	A	1
14.	B	1
15.	A	1
16.	D	1
17.	C	1
18.	D	1
19.	C	1
20.	B	1
21.	C	1
22.	B	1
23.	A	1
24.	A	1
25.	B	1

Section 2

Question			Expected response	Max mark	Additional guidance
1.	(a)	(i)	Nucleus	1	
		(ii)	(Site of) <u>photosynthesis</u>	1	
	(b)		It has no cell wall/vacuole	1	Also acceptable: no mitochondria
	(c)		1950	1	
2.	(a)		Any value between (+)11 and (+) 17	1	
	(b)		1	1	
	(c)		5.46/5.5	1	
3.	(a)	(i)	Substance J: Pyruvate (1) Substance K: Water (1)	2	
		(ii)	2	1	
	(b)		Mitochondrion/mitochondria	1	
	(c)		No oxygen/When oxygen is absent/ not enough oxygen/low levels of oxygen/low O ₂	1	Not acceptable: need/require oxygen
4.	(a)	(i)	Synthesis	1	
		(ii)	Glucose-1-phosphate/G1P	1	
	(b)	(i)	So that only starch made (during the experiment) would be detected. OR To prove that the starch present has been made (during the experiment).	1	Answer must imply the starch is being made in the experiment.
		(ii)	Volume of G1P/substrate Volume of (potato) extract/enzyme Volume of water Concentration of G1P/substrate Concentration of (potato) extract /enzyme Any 2	2	Not acceptable: – Mass/weight of potato(es) – Any mention of temperature – Any mention of pH
	(b)	(iii)	Starch is produced when the (potato) extract/enzyme is present or converse. There is no starch produced in 2 nd row/control.	1	Answer must refer to starch production rather than presence.

Question			Expected response	Max mark	Additional guidance
5.	(a)		2.6	1	
	(b)		37.5	1	
	(c)		1.8	1	
6.	(a)	(i)	Neuron Y = Sensory (neuron) (1) Neuron Z = Motor (neuron) (1)	2	
		(ii)	Chemicals	1	Also acceptable: neurotransmitters/ chemical messages Not acceptable: chemical messengers
	(b)		Muscle	1	Not acceptable: gland
	(c)		Protects (the body/hand/finger) from harm	1	Also acceptable: protects (the body/hand/finger) from danger
7.	(a)	(i)	Ovary/ovaries	1	
		(ii)	Sperm <u>nucleus</u> and egg <u>nucleus</u> fuse or join together/sperm and egg <u>nuclei</u> fuse/gamete <u>nuclei</u> fuse or join together	1	Must be clear it is the two nuclei which are involved.
		(iii)	Zygote	1	
	(b)		(Embryonic) stem (cells)	1	
8.	(a)	(i)	Affected Unaffected	1	
		(ii)	Heterozygous	1	
		(iii)	3	1	
	(b)		Male genotype rr Female genotype rr	1	
	(c)		Allele(s)	1	

Question			Expected response		Max mark	Additional guidance											
9.	(a)		Lymphocyte(s)		1												
	(b)	(i)	<table><tr><td colspan="3">Heading</td></tr><tr><td rowspan="2">Symptoms of long covid</td><td colspan="2">Percentage of patients (%)</td></tr><tr><td>Start of study</td><td>End of study</td></tr></table>		Heading			Symptoms of long covid	Percentage of patients (%)		Start of study	End of study	1	Not acceptable: symptoms alone/ symptoms of covid			
Heading																	
Symptoms of long covid	Percentage of patients (%)																
	Start of study	End of study															
		(ii)	<table><tr><td colspan="3">Relevant data</td></tr><tr><td>Extreme breathlessness</td><td>38</td><td>30</td></tr><tr><td>Memory and concentration issues</td><td>48</td><td>38</td></tr><tr><td>Other health issues</td><td>57</td><td>45</td></tr></table>		Relevant data			Extreme breathlessness	38	30	Memory and concentration issues	48	38	Other health issues	57	45	1
Relevant data																	
Extreme breathlessness	38	30															
Memory and concentration issues	48	38															
Other health issues	57	45															
	(c)		5 : 13 : 2		1												
	(d)		Too many variables to control/ difficult to control all variables/ symptoms could be caused by other factors		1	Accept correct example of a variable eg. age/ gender/ ethnicity/ previous underlying health issues/ lifestyle choices											

Question			Expected response	Max mark	Additional guidance
10.	(a)	(i)	Right ventricle	1	
		(ii)	Higher/more in N (than in L) OR Lower/less in L (than in N) OR High in N and low in L	1	Must be comparative Not acceptable: - L is oxygenated and N is deoxygenated - absolute terms eg no carbon dioxide.
	(b)	(i)	Coronary artery	1	
		(ii)	No oxygen/glucose/nutrients (1) No ATP produced/no energy released (1)	2	
11.	(a)		- Large surface area - Thin walls - Extensive blood supply or suitable description Any 2	2	Also acceptable: dense capillary network or suitable description Not acceptable: - Thin cell walls - Thin on its own
	(b)		As the temperature (of the water) increases (average) breathing rate increases OR As the temperature (of the water) decreases (average) breathing rate decreases	1	Must show correct causation Must refer to rate/breaths per minute and not just breaths
12.	(a)	(i)	32	1	
		(ii)	Carry out more transects/use more sites (along the transect)/use more quadrats Repeat the investigation/ experiment	1	Not acceptable: repeat (it)
	(b)	(i)	Indicator (species)	1	
		(ii)	Abiotic (factors)	1	

Question			Expected response	Max mark	Additional guidance
13.	(a)	(i)	As carbon dioxide (concentration) increases the (mass of) sugar produced increases until 0.08(%) (1) then levels off (1)	2	Must show correct causation to access any marks 1 mark for correct description of increase then levels off [without mention of 0.08(%)]
		(ii)	Temperature OR Light intensity	1	Not acceptable: light on its own
	(b)	(i)	Splitting of water/water is split	1	
		(ii)	Substance: Cellulose (1) Role: Structural/to make (plant) cell wall/component of cell wall (1)	2	
14.			Role of pond weed 1. Pond weed is a producer/makes its own food/carries out photosynthesis OR Pond weed is a source of energy/food for the tadpole/ (primary) consumer Energy at each level 2. Pond weed/producer converts light energy into chemical energy 3. Energy is transferred/passed on from one level to the next 4. Only a (very) small quantity is used for growth AND is available at the next level 5. Majority of energy is lost 6. as heat/movement/undigested materials/faeces 7. Energy decreases/is lost (at each level in the food chain) OR Pond weed has the most energy AND pike has the least energy	4	Also acceptable: energy is transferred/passed on from one named organism to another ≤ 10% = small ≥ 90% = majority

Question			Expected response	Max mark	Additional guidance																	
15.	(a)		Nitrate(s)	1	Also acceptable: nitrogen, potassium, phosphorous or any other correct example																	
	(b)	(i)	Y-axis scale and label including units (1) All disease bars at correct height (1) <table border="1"><thead><tr><th rowspan="2">Crop</th><th colspan="2">Average loss in yield (%)</th></tr><tr><th>Insects</th><th>Disease</th></tr></thead><tbody><tr><td>Wheat</td><td>4</td><td>7</td></tr><tr><td>Barley</td><td>7</td><td>6</td></tr><tr><td>Oats</td><td>8</td><td>14</td></tr><tr><td>Oil seed rape</td><td>8</td><td>12</td></tr></tbody></table>	Crop	Average loss in yield (%)		Insects	Disease	Wheat	4	7	Barley	7	6	Oats	8	14	Oil seed rape	8	12	2	Scale - any three values to establish a linear scale
Crop	Average loss in yield (%)																					
	Insects	Disease																				
Wheat	4	7																				
Barley	7	6																				
Oats	8	14																				
Oil seed rape	8	12																				
		(ii)	Wheat	1																		
		(iii)	Barley shows a greater loss to insects than disease	1	Also acceptable: – Barley is affected more by insects than disease																	
		(iv)	175 000	1																		
	(c)		Bioaccumulation	1																		
	(d)		Biological control	1																		
16.	(a)		Geographical/Ecological/Behavioural	1	Named example alone is not acceptable																	
	(b)	(i)	DNA	1																		
		(ii)	Radiation OR Chemicals	1	Also acceptable: named radiation Also acceptable: named chemical																	
	(c)		Competition for food OR (Type/source of) food available	1																		
	(d)		(A group of) organisms that can interbreed to produce fertile offspring.	1	Answer must imply • there is more than one organism • breed/mate/reproduce • production of fertile offspring																	

[END OF MARKING INSTRUCTIONS]

General marking principles for National 5 Biology

This information is provided to help you understand the general principles you must apply when marking candidate responses to questions in this paper. These principles must be read in conjunction with the marking instructions, which identify the key features required in candidate responses.

- (a) Marks for each candidate response must **always** be assigned in line with these general marking principles and the detailed marking instructions for this assessment.
- (b) Marking should always be positive. This means that, for each candidate response, marks are accumulated for the demonstration of relevant skills, knowledge and understanding: they are not deducted from a maximum on the basis of errors or omissions.
- (c) If a specific candidate response does not seem to be covered by either the principles or detailed marking instructions, and you are uncertain how to assess it, you must seek guidance from your team leader.
- (d) There are no half marks awarded.
- (e) Where a candidate makes an error at an early stage in the first part of a question, credit should normally be given for subsequent answers that are correct with regard to this original error. Candidates should not be penalised more than once for the same error.
- (f) Unless a numerical question specifically requires evidence of working to be shown, full marks should be awarded for a correct final answer (including units, if appropriate) on its own.
- (g) In the detailed marking instructions, if a word is underlined then it is essential; if a word is (bracketed) then it is not essential.
- (h) In the detailed marking instructions, words separated by / are alternatives.
- (i) A correct answer can be negated if:
 - an extra, incorrect, response is given
 - additional information that contradicts the correct response is included.
- (j) Unless otherwise required by the question, use of abbreviations (eg DNA, ATP) or chemical formulae (eg CO₂, H₂O) are acceptable alternatives to naming.
- (k) Where incorrect spelling is given:
 - If the correct word is recognisable then give the mark.
 - If the word can easily be confused with another biological term then do not give the mark eg mitosis and meiosis.
 - If the word is a mixture of other biological words then do not give the mark, eg osmotis, respirduction, protosynthesis.
- (l) Presentation of data
 - If a candidate provides two graphs or charts, mark both and give the higher score.
 - If a question asks for a particular type of graph and the wrong type is given, then full marks cannot be awarded. Candidates cannot achieve the plot mark but **may** be able to achieve the mark for scale and label.
 - If the x and y data are transposed, then do not give the scale and label mark.
 - If the graph uses less than 50% of the axes, then do not give the scale and label mark.
 - If 0 is plotted when no data is given, then do not give the plot mark (ie candidates should only plot the data given).
 - No distinction is made between bar graphs and histograms for marking purposes.
 - In a pie chart lines must originate from the central point and extend to tick marks. Labels must be given in full.

- (m) Marks are awarded only for a valid response to the question asked. For example, in response to questions that ask candidates to:
- **identify, name, give or state**, they need only answer or present in brief form;
 - **describe**, they must provide a statement as opposed to simply one word;
 - **explain**, they must provide a reason for the information given;
 - **compare**, they must demonstrate knowledge and understanding of the similarities and/or differences between topics being examined;
 - **calculate**, they must determine a number from given facts, figures or information;
 - **predict**, they must indicate what may happen based on available information;
 - **suggest**, they must apply their knowledge and understanding to a new situation.

2024 Nat5 Biology Section 1 Multiple Choice Analysis

Item	Key	A	B	C	D	Missing	Invalid
1	C	14	13	65	6	1	0
2	C	11	39	44	5	1	0
3	D	18	23	13	45	1	0
4	A	53	26	10	10	2	0
5	C	19	56	17	7	1	0
6	D	5	3	9	82	1	0
7	B	7	50	26	15	1	0
8	B	20	43	3	32	1	0
9	C	15	12	64	7	1	0
10	A	72	6	12	9	1	0
11	D	24	17	11	47	1	0
12	D	4	2	12	81	1	0
13	A	44	15	32	7	1	0
14	B	12	63	9	15	1	0
15	A	66	11	13	9	1	0
16	D	16	13	21	48	1	0
17	C	18	17	59	4	1	0
18	D	13	11	10	65	1	0
19	C	2	5	90	1	1	0
20	B	10	80	5	4	1	0
21	C	6	11	74	7	1	0
22	B	13	60	20	6	1	0
23	A	26	24	47	2	1	0
24	A	88	3	3	4	1	0
25	B	2	55	38	3	1	0

2024 Nat5 Biology Section 2 Marks Data Analysis

2024 Item	Max Mark	Average Mark	% Average Mark	Discrimination	% Not Attempted
Q1ai	1	0.9	86.2%	7	0.3%
Q1aai	1	0.7	66.3%	49	4.9%
Q1b	1	0.7	72.8%	36	2.5%
Q1c	1	0.6	57.4%	43	2.6%
Q2a	1	0.8	83.7%	34	4.1%
Q2b	1	0.6	60.1%	50	2.7%
Q2c	1	0.4	37.2%	54	11.3%
Q3ai	2	1.1	55.6%	63	2.0%
Q3aai	1	0.8	75.0%	41	3.9%
Q3b	1	0.8	76.2%	46	3.4%
Q3c	1	0.4	38.3%	53	9.6%
Q4ai	1	0.5	53.5%	64	9.2%
Q4aai	1	0.5	52.6%	59	6.2%
Q4bi	1	0.1	7.8%	20	7.1%
Q4bii	2	0.6	31.1%	34	3.3%
Q4biii	1	0.2	17.8%	32	10.2%
Q5a	1	0.7	73.5%	38	0.9%
Q5b	1	0.3	28.8%	46	5.6%
Q5c	1	0.5	53.5%	54	5.4%
Q6ai	2	1.4	70.3%	58	6.0%
Q6aai	1	0.3	29.1%	46	9.2%
Q6b	1	0.5	46.2%	51	8.3%
Q6c	1	0.3	35.0%	36	4.1%
Q7ai	1	0.9	91.2%	27	1.0%
Q7aai	1	0.3	26.1%	49	2.9%
Q7aaii	1	0.7	70.3%	57	5.2%
Q7b	1	0.6	64.8%	57	11.1%
Q8ai	1	0.6	61.0%	46	2.9%
Q8aai	1	0.4	37.2%	53	7.3%
Q8aaii	1	0.5	47.9%	34	1.0%
Q8b	1	0.5	52.1%	55	3.5%
Q8c	1	0.5	49.8%	58	9.1%
Q9a	1	0.5	50.2%	65	13.1%
Q9bi	1	0.4	44.9%	27	7.0%
Q9bii	1	1	95.9%	20	1.2%
Q9c	1	0.7	66.7%	52	5.4%
Q9d	1	0.2	22.8%	24	7.0%
Q10ai	1	0.7	67.9%	53	3.6%
Q10aai	1	0.5	49.0%	50	7.1%
Q10bi	1	0.3	31.6%	58	5.6%
Q10bii	2	0.3	17.3%	42	6.3%
Q11a	2	0.9	42.7%	68	7.2%
Q11b	1	0.9	85.8%	31	2.3%
Q12ai	1	0.8	82.5%	38	3.2%
Q12aai	1	0.5	51.5%	28	3.4%
Q12bi	1	0.6	63.3%	63	9.8%
Q12bii	1	0.4	41.7%	50	3.0%
Q13ai	2	0.7	36.3%	51	5.9%
Q13aai	1	0.7	66.8%	47	3.8%
Q13bi	1	0.3	30.8%	53	17.5%
Q13bii	2	0.7	32.6%	61	7.7%
Q14	4	2.1	53.3%	64	2.9%
Q15a	1	0.6	64.2%	62	9.8%
Q15bi	2	1.4	70.1%	43	4.5%
Q15bii	1	0.8	80.8%	26	2.0%
Q15biii	1	0.4	41.2%	42	7.9%
Q15biv	1	0.3	25.3%	47	10.4%
Q15c	1	0.4	37.5%	62	23.7%
Q15d	1	0.4	44.9%	61	14.3%
Q16a	1	0.4	40.8%	52	8.3%
Q16bi	1	0.3	28.2%	37	10.6%
Q16bii	1	0.4	43.5%	61	5.8%
Q16c	1	0.2	18.1%	34	15.7%
Q16d	1	0.4	40.1%	58	3.6%
Paper	75	37.9	50.6%		

2024 Nat5 Biology Section 2 Distribution of Marks Analysis

2024 Item	0	1	2	3	4
Q1ai	14	86			
Q1aii	34	66			
Q1b	27	73			
Q1c	43	57			
Q2a	16	84			
Q2b	40	60			
Q2c	63	37			
Q3ai	29	30	41		
Q3aii	25	75			
Q3b	24	76			
Q3c	62	38			
Q4ai	46	54			
Q4aii	47	53			
Q4bi	92	8			
Q4bii	52	35	14		
Q4biii	82	18			
Q5a	26	74			
Q5b	71	29			
Q5c	47	53			
Q6ai	19	21	60		
Q6aii	71	29			
Q6b	54	46			
Q6c	65	35			
Q7ai	9	91			
Q7aii	74	26			
Q7aiii	30	70			
Q7b	35	65			
Q8ai	39	61			
Q8aii	63	37			
Q8aiii	52	48			
Q8b	48	52			
Q8c	50	50			
Q9a	50	50			
Q9bi	55	45			
Q9bii	4	96			
Q9c	33	67			
Q9d	77	23			
Q10ai	32	68			
Q10aii	51	49			
Q10bi	68	32			
Q10bii	70	24	5		
Q11a	43	29	28		
Q11b	14	86			
Q12ai	18	82			
Q12aii	49	51			
Q12bi	37	63			
Q12bii	58	42			
Q13ai	54	20	26		
Q13aii	33	67			
Q13bi	69	31			
Q13bii	62	11	27		
Q14	8	19	35	27	11
Q15a	36	64			
Q15bi	16	28	56		
Q15bii	19	81			
Q15biii	59	41			
Q15biv	75	25			
Q15c	63	37			
Q15d	55	45			
Q16a	59	41			
Q16bi	72	28			
Q16bii	57	43			
Q16c	82	18			
Q16d	60	40			

2024 Nat5 Biology Section 3 Assignment Marks Analysis

Item	Max Mark	Average Mark	% Average Mark	Discrimination
1	1	0.8	84.7%	29
2	3	1.5	48.6%	45
3a	1	0.6	60.2%	30
3b	1	1	96.0%	17
3c	1	0.5	52.0%	39
3d	1	0.7	66.4%	33
3e	1	0.7	68.6%	26
3f	1	0.9	87.0%	28
4a	1	0.9	94.4%	27
4b	1	0.7	73.3%	29
4c	1	0.8	79.2%	37
4d	1	0.6	62.4%	36
5	1	0.2	17.8%	34
6	1	0.2	20.2%	32
7	2	0.9	44.6%	41
8a	1	0.8	80.5%	20
8b	1	1	96.7%	23
Paper	20	12.7	63.7%	

Item	Max Mark	0	1	2	3
1	1	15	85		
2	3	26	27	23	24
3a	1	40	60		
3b	1	4	96		
3c	1	48	52		
3d	1	34	66		
3e	1	31	69		
3f	1	13	87		
4a	1	6	94		
4b	1	27	73		
4c	1	21	79		
4d	1	38	62		
5	1	82	18		
6	1	80	20		
7	2	42	26	32	
8a	1	19	81		
8b	1	3	97		

Item	Max Mark	Item Description
1	1	An aim that describes clearly the purpose of the investigation.
2	3	An account of biology relevant to the aim of the investigation.
3a	1	Brief description of approach used to collect experimental/fieldwork data.
3b	1	Sufficient raw data from the candidate's experiment/fieldwork.
3c	1	Data presented in a correctly produced table
3d	1	Mean and/or derived values calculated correctly
3e	1	Data/information relevant to the aim from an internet/literature source.
3f	1	A reference for the source of the internet/literature data or information.
4a	1	An appropriate format from the options of bar graph or line graph.
4b	1	The axes of the graph have suitable scales.
4c	1	The axes of the graph have suitable labels and units
4d	1	Accurately plotted data points and a line (line graph) or clear bar tops (bar graph).
5	1	A valid comparison of the experimental/fieldwork data with data/information from the internet/literature source.
6	1	A conclusion that relates to aim and supported by all the data in the report.
7	2	An evaluation of the experimental/fieldwork procedure.
8a	1	Informative Title
8b	1	Clear and Concise Report